

5.1.10 Product Notation

The notation for the product of a sequence of numbers is analogous to the notation for their sum. The greek capital letter pi, $\prod$, denotes a product. For example,

$$\prod_{k=1}^5 a_k = a_1 a_2 a_3 a_4 a_5$$

Definition

If m and n are integers and $m \leq n$, the symbol $\prod_{k=m}^n a_k$, read the **product from k equals m to n of a-sub-k**, is the product of all the terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$. We write

$$\prod_{k=m}^n a_k = a_m \times a_{m+1} \times a_{m+2} \times \dots \times a_n$$

A recursive definition for the product notation is the following: If m is any integer, then

$$\prod_{k=m}^m a_k = a_m \text{ and } \prod_{k=m}^n a_k = \left(\prod_{k=m}^{n-1} a_k \right) \times a_n \text{ for every integer } n > m$$